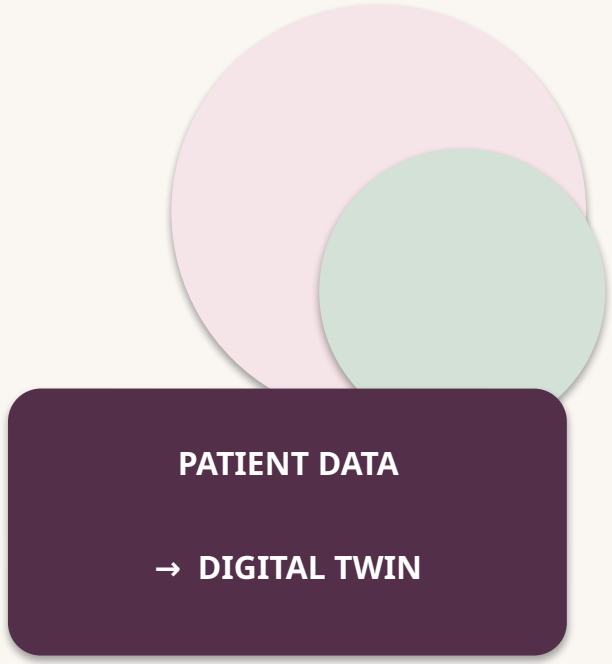
PROJECT PRESENTATION | 2026

NAARI

A PCOS-focused Digital Twin research and decision-support prototype



Connecting clinical history, menstrual records, symptoms, simulated wearable data and short-term glucose prediction.

Team: NAARI • College: BVRIT Hyderabad College of Engineering for Women

Research prototype • Synthetic demo data • Not clinically validated

Team members: Namratha Raj Goud Palpanoor (Lead) • Priyanka K • Samskruthi Batikiri • Sarayu P

Patient information is often fragmented

NAARI explores a way to review relevant information together over time.

Clinical history

Lab reports, verified measurements and relevant medical history.

Cycle & symptoms

Recorded period starts, cycle intervals and patient-reported symptoms.

Time-series data

Simulated glucose, heart rate and step readings across time.

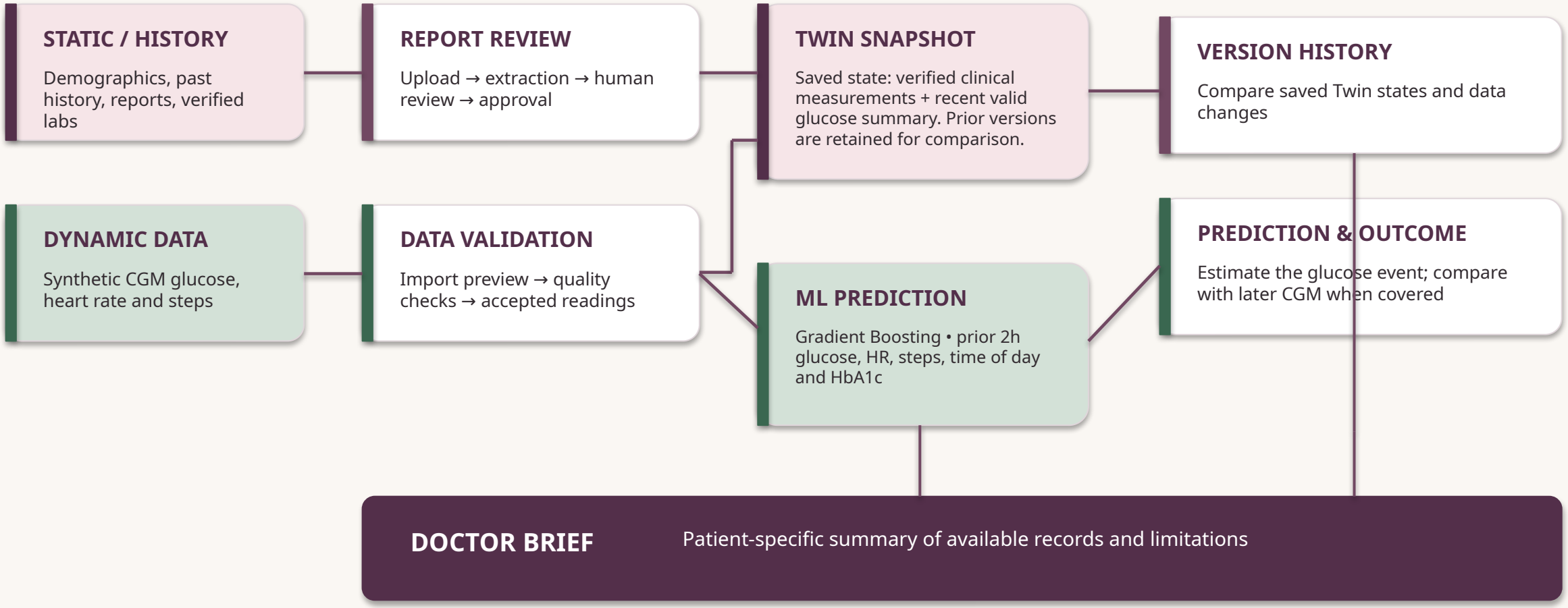
NAARI's focused prediction task

INPUT WINDOW	MODEL OUTPUT	PREDICTION HORIZON
Previous 2 hours	**Glucose >180 mg/dL**	**Next 120 minutes**

The model runs only when current glucose is at or below 180 mg/dL. It is experimental and trained on synthetic data.

How NAARI's data flows through the prototype

Two data streams are reviewed and processed, then used by separate Twin and prediction components.



Boundary: cycle and symptom records remain separate in this release; HR and steps are model inputs but are not stored directly in the saved Twin state.

A versioned summary, not just a dashboard

The application keeps source records, the saved Twin state and prediction inputs distinct.

RECORDS

Reports • verified lab values • menstrual history • symptoms • wearable imports

TWIN SNAPSHOT

Eligible verified clinical values + recent valid glucose summary

VERSION COMPARISON

Inspect saved versions and identify what changed

What the Twin keeps

- Verified clinical measurements
- Recent valid glucose-data summary
- Dated, versioned snapshots for comparison

What remains separate

- Menstrual-cycle and symptom history
- Heart rate and steps used as prediction features
- Raw/source records that can be inspected separately

A saved Twin update is distinct from running a prediction.

A narrow, clearly defined prediction task

The current model demonstrates integration of a trained ML model into the application; it is not a PCOS diagnostic model.

FEATURES

Recent glucose summary • heart-rate mean • steps • time of day • latest verified HbA1c

MODEL

GradientBoostingClassifier (scikit-learn), trained on synthetic data

OUTPUT

Estimated likelihood of glucose >180 mg/dL in the next 120 minutes, when current glucose ≤180

Evaluation shown in the app

- Gradient Boosting compared with a logistic-regression baseline
- Metrics include AUROC, AUPRC, recall, precision and Brier score
- Grouped patient split in the training/evaluation workflow

Interpretation boundary

- Synthetic training data; no clinical validation
- A model score is not a confirmed medical risk
- Does not predict PCOS diagnosis, cycle date, ovulation or fertility

The technical objective is a glucose-event prediction, not a claim of clinical readiness.

What the reviewer can see and test

The demo follows one detailed patient, then checks how the app handles limited or empty records.

01	**Select patient**	Ananya / Meera / Kavya; distinct data availability
02	**Review a report**	Upload or inspect sample PDF; review extracted fields before approval
03	**Explore records**	Lab tests, cycle timeline, symptom severity and wearable time-series
04	**Import later data**	Validate a synthetic CGM batch and update the saved Twin
05	**Inspect model**	Run prediction when ready; show input features and evaluation
06	**Generate brief**	Patient-specific Doctor Brief for clinical review / PDF export

A working research prototype with explicit boundaries

NAARI demonstrates a data-to-model workflow while keeping synthetic evidence and clinical readiness separate.

WHAT THE PROTOTYPE DEMONSTRATES

Report review and approval • imported time-series validation • versioned Twin snapshots • glucose-event prediction • patient-specific Doctor Brief

CURRENT LIMITATIONS

Synthetic demo data • no live sensor connection • prediction not clinically validated • no PCOS diagnostic or fertility prediction

Next research steps

- Evaluate with suitable, ethically permitted and properly governed research data.
- Assess model generalisation, calibration and clinical usefulness before real-world use.
- Explore reliable sensor integration and prospective validation only after the appropriate safeguards are in place.

NAARI is a clinical decision-support research prototype, not a diagnostic or treatment system.